GEORGE LARSON

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Updated 2026-04 Director of Technology with 25+ years building and operating software, infrastructure, and manufacturing systems. I have led teams at Sony Electronics, scaled a medical billing startup to national reach, programmed PLCs on production lines, and directed technology for a mid-size manufacturer. I do Linux platforms, DevOps automation, ecommerce, and security programs. I stay close to the keyboard even when the job is strategic. I build AI tooling that actually ships — multi-agent infrastructure, MCP servers, model-routing pipelines, and SDKs in the Venice AI ecosystem.

KEY METRICS

- 36 public assets scanned continuously via HostedScan + OWASP ZAP with 33 findings closed in the past 60 days and zero critical/high issues in backlog.
- 36 active GitLab projects (11 users, 2 groups), 2,204 merge requests, 739 CI pipelines (avg 565 successes / 58 fails) and 16 managed SSH keys.
- Mattermost workspace with 7 teams, 40 channels, ~1.0M lifetime posts, 313 calls, and alert channels wired to uptime, upgrade, helpdesk, GitLab, and web-critical events.
- Zammad helpdesk processed 1,487 tickets YTD (~150–206/month in Jun–Aug) with ≥97% close rate, 68% resolved inside 24 hours (56% inside 8 hours), zero reopen events.
- Uptime Kuma monitors 31 services (GitLab, Helpdesk, Shopify APIs, FTP, chat, docs, etc.) as part of 24×7 observability feeding Mattermost.

CURRENT FOCUS

- Seeking a full-time technology leadership role where I can build, ship, and lead teams hands-on.
- Operating a multi-machine agent mesh (Agent Zero with A2A routing across nodes; nullclaw on a hardened perimeter; ironclaw as private agent; automated job-application pipeline as a working production case study) with tiered inference and clear security boundaries.
- Building in the Venice AI SDK ecosystem — TypeScript, PHP, and Rust SDKs; an MCP server; an AI research tool (deep-research-privacy, 35★) — alongside continued systems work in Rust and Go (netgrep, termshark, wiregraph).
- Running larson.help, a small AI consulting practice, in parallel with the job hunt.
- Strengthening security programs, compliance boundaries, and knowledge management systems that keep data and people protected.

CORE STRENGTHS

Software Development and Engineering

- Full stack delivery across Python, Go, Rust, C#, TypeScript, PHP, Perl, C++, SQL.
- Code review, automated testing, and performance tuning for legacy and greenfield systems.

DevOps and Infrastructure Automation

- CI and CD design, infrastructure as code, container orchestration, and repeatable releases on self-hosted runners.
- Observability, incident response, high-availability design (replication, backups, CDN), and pragmatic SLO tracking for hybrid environments.

Technology Leadership and Strategy

- Strategic roadmaps that connect executive goals to team-level delivery.
- Calm stewardship for blended internal, offshore, and vendor teams.

Security and Privacy

- Network hardening, IAM, HostedScan/ZAP/Nikto assessment loops, and zero-trust rollouts.
- Regulated-data guardrails and tabletop response drills.

Data Engineering and Automation

- OCR digitization, ETL pipelines, semantic search, and knowledge-base design.
- Workflow automation using scripting, open models, and RAG pipelines where they deliver measurable ROI.

OPEN SOURCE PROJECTS

Venice AI Ecosystem (2025 to Present)

- deep-research-privacy (35 stars, TypeScript) — Privacy-first AI research tool using Venice's uncensored mode; iterative search and synthesis pipeline that runs end-to-end against a local or self-hosted endpoint.
- venice-dev-tools (14 stars, TypeScript) — Comprehensive SDK for the Venice AI API; widely-installed npm package powering downstream tools.
- venice-ai-php (7 stars, PHP) — Drop-in OpenAI-compatible PHP SDK for Venice.
- venicecode (5 stars, TypeScript) — AI-powered development tool integrating Venice into the editor workflow.
- venice-mcp (4 stars, TypeScript) — Model Context Protocol server exposing Venice models to MCP-aware clients (Claude Desktop, Cline, Cursor).
- venice-ai-api-sdk-rust, venice-ai-client, venice-cache-monitor — Rust SDK, lightweight TypeScript client, and a cache-behavior test harness.

LLM Showdown (2026 to Present)

- llm-showdown-data — Companion dataset for the blog series benchmarking 15+ free coding models on a $25 VPS using opencode and a ralph-wiggum-style iterative loop.
- Designed and runs the evaluation harness; publishes per-round results comparing model behavior on real coding tasks.

Job Search Agent (2026 to Present)

- Multi-machine agent pipeline: laptop scraper feeds an Agent Zero task on a remote node that scores LinkedIn and Built In job postings against a structured profile, drafts cover letters under voice and revision rules, and emits a structured queue.
- Companion laptop processor mutates the application tracker, builds PDFs, commits to git, and posts a Mattermost digest. Inbox-sweep classifier reads Gmail replies via Composio and routes rejections through the same tracker convention.
- Demonstrates the production shape — source adapter, scoring profile, draft+scrub pass, audited tracker — that generalizes to client lead-monitoring agents.

Fracture (2 stars, TypeScript) — Modernization Case Study (2026)

- Modernizes Mozilla's 2012 BrowserQuest into a contemporary multiplayer architecture with eight integrated AI services: RAG, semantic search, agent-driven triage, and content moderation.
- Live demo at fracture.georgelarson.me. The case study itself is the deliverable: how to take a frozen reference codebase from a major vendor and bring it forward without rewriting from scratch.

nullclaw + ironclaw — Agent Mesh (Zig + Rust, 2026 to Present)

- nullclaw: public-facing AI concierge ("Nully") on a hardened perimeter box. Tiered inference (Haiku for conversational triage, Sonnet for code analysis). Full self-hosted comms stack (Ergo IRC, gamja web client, Cloudflare-proxied WebSocket, Let's Encrypt TLS).
- ironclaw: private personal agent on a separate system with distinct security boundaries. Agent-to-agent routing between the two.
- Now extended into a multi-node Agent Zero mesh with A2A routing, dedicated chat and utility models, and per-node project isolation.

netgrep (Rust), Creator and Maintainer (2026 to Present)

- Modern replacement for ngrep: live packet capture and pcap analysis with TCP stream reassembly, TLS 1.2/1.3 decryption, HTTP/2 and DNS parsing, and an interactive terminal UI.
- Maintains ngrep-compatible CLI flags; outputs colored text, JSON, hex dump, or pcap.

termshark (Go), Contributor and Maintainer (2026 to Present)

- Modernized fork of termshark, a terminal UI for Wireshark/tshark.
- Refactored architecture, improved test coverage, updated to Go 1.22+ idioms, and added an experimental browser-based web interface.

wiregraph (Rust, 2026 to Present)

- GPU-rendered network traffic visualizer. Hosts as nodes, flows as edges, real-time topology view of a live capture session.

WORK EXPERIENCE

Director of Technology, FM Expressions (2018 to Present)

- Led technology strategy for a mid-size manufacturing company serving ecommerce channels.
- Improved database performance by 93 percent while driving critical systems to 99 percent uptime; cut MySQL slow-query volume by 96 percent to accelerate incident response.
- Built a knowledge management platform (Wiki.js, Gitea, Qdrant, ETL pipelines) that captures production logs, transcripts, and operator notes so teams can search, surface, and act on institutional knowledge in real time.
- Resolved conflicting manufacturing documentation, created a style guide and glossary, and instituted monthly and quarterly review cycles to keep safety and operations guidance consistent.
- Reduced a 1.03 GB legacy software repository (4,656 binary artifacts) to a clean, LFS-ready codebase, eliminating clone failures and enabling sustainable development practices.
- Hardened hybrid infrastructure with Authentik SSO, HashiCorp Vault, automated TLS renewal, firewall hardening (iptables, UFW, CrowdSec, fail2ban), DMARC/SPF/DKIM enforcement, HostedScan/OWASP ZAP sweeps across 36 public assets, and Red Sift OnDMARC (p=reject, 1.5M emails/month at 99.48% compliance) to keep critical/high findings at zero and close 33 issues in the past 60 days.
- Run GitLab (36 projects, 11 users, 739 CI pipelines) and Mattermost (7 teams, 40 channels, ~1M posts) as the collaboration core tied into Redmine (22 cross-functional projects), Zammad (~200 tickets/month with 68% closed inside 24 hours), and Uptime Kuma (31 monitors feeding alert channels).
- Maintained high-availability platforms: dual MySQL replicas, VMware ESXi with Datto backups, Active Directory, Cisco Meraki switching, SonicWall/pfSense firewalls, and a staged CDN migration from self-hosted edges to Contabo and Cloudflare.
- Built the Mashy internal dashboard plus tooling such as Psono vault, PrivateBin, phpMyAdmin/Adminer, DBeaver, knowledge mining utilities, and status pages so teams access credentials, data stores, and observability in one click.
- Driving adoption metrics for the knowledge stack to quantify time-to-answer gains for production and customer operations.
- Captured Meteor print-head PCAPs and tuned Nikto, Metasploit, and custom fuzz rigs to secure industrial protocols while migrating remote access from VPN to zero-trust policies.
- Implemented Agile delivery, ticketing, and analytics to align product, operations, and finance on throughput, defect, and margin targets.

Software Developer (2017)

- Departed after company practices conflicted with professional engineering standards.

Senior Software Engineer and Manager, Universal SmartComp (2009 to 2016)

- Helped scale the company from a 50 person startup to a national leader in medical billing.
- Designed EDI frameworks, ACH processing, ERP integrations, and a medical payer portal.
- Led internal and offshore teams across development and Linux infrastructure.
- Acted as security response lead and Linux SME; migrated project management to Redmine with custom plugins.
- Built OCR bill entry validation that reduced errors and sped turnaround.

Software Engineer and Team Manager, Sony Electronics (2004 to 2009)

- Directed software development, logistics, and technical operations for the Pittsburgh Customer Satisfaction Center.
- Designed and ran the TiVo firmware upgrade station for the United States market.
- Built hardware testing suites for CD drives, set-top boxes, PCBs, televisions, cameras, and MP3 players.
- Created an AS/400 parts ordering interface that saved millions in operational spend.
- Managed more than 50 combined technical and non-technical staff.
- Developed a VT-52 terminal emulation system used for inventory management.

Maintenance and Automation Technician, T. Marzetti Company (2000 to 2004)

- Led second shift maintenance for robotics, conveyors, and packaging machinery.
- Programmed PLCs and tuned sensors, actuators, and industrial controls on the production line.

Earlier Career Highlights (1990s to 2000s)

- Software developer and webmaster for non-profit organizations.

- Engineer for The Sounding Board, a psychotherapeutic BBS in Columbus, Ohio; built custom C++ modules for Major BBS and kept one of the first internet-connected BBS systems alive.
- Contract developer for Daifuku USA working on material handling support systems.
- Two-time Business Professionals of America chapter president; regional competitor in extemporaneous speech.

VOLUNTEER WORK

- Systems administrator for a volunteer Linux community, digital radio station, and open shell environment.
- Long-term support for animal shelters, including events, outreach, and facilities work.
- OCR and digitization project contributor for historical legal documents.

TECHNICAL SKILLS

Languages: Python, Go, Rust, Zig, C#, TypeScript, PHP, Perl, C++, Bash, SQL

Web and Data: MySQL, PostgreSQL, WordPress, Redmine, Gitea, Wiki.js

Infrastructure: Linux, Apache, Nginx, Caddy, IIS, Kubernetes, VMware ESXi, Datto backups, Cisco Meraki, SonicWall, pfSense, CrowdSec, Let's Encrypt/ACME, Cloudflare CDN

Cloud: AWS, Azure, Google Cloud, Proxmox

Security and Identity: Authentik, HashiCorp Vault, Psono, CrowdSec, HostedScan, OWASP ZAP, Nikto, Metasploit, SSL/TLS, iptables, UFW, fail2ban, DMARC/SPF/DKIM, DefectDojo, OpenVAS, zero-trust network design

Collaboration: Agile, Scrum, Kanban, Redmine, Zammad, GitLab, Mattermost, CryptPad

Mail Systems: iRedMail, Postfix, Dovecot

Commerce: Shopify, custom ecommerce platforms, payment integrations

Data and Automation: RAG design, OCR pipelines, semantic search, ETL, knowledge base design

AI Agent Infrastructure: Agent Zero (A2A protocol, multi-node mesh, project isolation), Model Context Protocol (MCP server design), Composio toolkit integration, Venice AI SDK ecosystem (TypeScript/PHP/Rust SDKs), LiteLLM, multi-provider model routing (Anthropic, Z.AI GLM, Venice, MiniMax, Groq, Featherless), tiered inference (utility/chat split), prompt caching, agent-driven CI and content pipelines, RAG and semantic search, agent security boundaries, scheduled autonomous tasks

Systems and Networking: TCP/IP analysis, packet capture, PLC programming, industrial controls, IRC server administration

Monitoring & Dashboards: Uptime Kuma, HostedScan, Red Sift OnDMARC, Dashy internal portal

EDUCATION AND PROFESSIONAL DEVELOPMENT

- Self-taught from C++ books as a teenager; built BBS modules in production before attending college.
- Continuous learner across hundreds of online courses, lectures, and certifications (see LinkedIn for details).
- Currently studying: Stanford CS336 — Language Modeling from Scratch (2025).
- Additional coursework spanning systems engineering, network security, database internals, and cloud architecture through platforms including Coursera, MIT OpenCourseWare, and vendor certification programs.